Pivotal Tokens Are Linearly Predictable Before They Are Generated

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Abstract

Pivotal Token Search (PTS) labels a generated token as pivotal when it moves the model's probability of reaching a correct answer past a threshold, estimated by sampling many rollouts and adjudicating each with an oracle. We ask whether that expensive, outcome-grounded property is already present in the model's activations *one position before* the token is emitted — the only position at which a prediction could still gate an intervention. We generate PTS events for three model scales (Qwen3-0.6B, 1.7B, 4B) over 3,000 GSM8K questions each, using a reimplementation of the search that batches rollouts across queries and emits exact token positions, and train linear probes at $t-1$. Pivotality is decodable at every scale, and the probe beats every cheaper explanation we tested — token identity, next-token entropy, top-2 margin, top-1 probability — with differences bootstrapped over questions on paired resamples. The informative pattern is how the margin *scales*: probe AUROC rises monotonically with model size ($0.807 \rightarrow 0.863 \rightarrow 0.897$) while entropy stays flat ($0.693 \rightarrow 0.721 \rightarrow 0.694$), so the advantage over predictive uncertainty widens from $+0.112$ (p=0.013) to $+0.204$ (p<0.001). Larger models encode pivotality more linearly, and what they encode is increasingly *not* uncertainty. We also show the comparison is fragile in a way that is easy to miss: drawing negative examples from the whole rollout rather than the span where pivots occur inflates entropy to 0.814 and erases the advantage entirely. Moving the probe to the pivot token itself inverts entropy to *below chance*, replicated across scales — uncertainty peaks immediately before a pivotal token and collapses immediately after. We further find that the *sign* of the impending shift — whether the next token will help or hurt — is itself decodable at $t-1$, but only at scale: a clean null at 0.6B (separated from nothing, not even a random direction) becoming 0.780 AUROC at 4B, separated from every cheaper explanation. Because entropy is sign-blind, and because a lookup on the current token cannot encode the sign of a token not yet emitted, this result is immune to both deflationary accounts. We also document two ways such measurements inflate: collapsing a question's sampled rollouts to one canonical sequence discards 74% of the supervision, and selecting the best layer on the reporting split adds 0.063 AUROC.

1 Introduction

A language model solving a multi-step problem does not distribute its risk evenly across the tokens it emits. The Phi-4 technical report [Abdin et al., 2024] formalized this as *Pivotal Token Search* (PTS): a token t is pivotal when appending it to a prefix moves the estimated probability of eventually solving the task past a threshold,

$$|\Delta p| = \big|P(\text{success} \mid \text{prefix} + t) - P(\text{success} \mid \text{prefix})\big| > \tau. \tag{1}$$

Estimating either term requires sampling many completions and adjudicating each with an oracle, so PTS is an offline instrument that costs tens of rollouts per candidate position.

Our starting question was whether that instrument can be amortized: is pivotality linearly decodable from the residual stream at $t-1$, before the token is produced? A probe read one step early is a different object from one read at the pivotal token itself—the latter is a post-hoc classifier, the former is available while there is still a decision to influence.

It is. But the more informative result is *what else* predicts it. The question is only interesting if the answer is not already known for a cheaper reason, and two cheaper reasons exist. *Token identity*: some tokens may simply be pivotal more often, in which case a lookup table suffices and no representation is implicated. *Next-token uncertainty*: high-entropy "forking" tokens are the incumbent notion of a critical position [Wang et al., 2025].

The probe clears both, but the margin over uncertainty is small enough that how the comparison is set up decides the answer. We show that an unexamined choice about where negative examples are drawn from can erase it entirely, and we report the comparison both ways.

Our contributions:

- Evidence that pivotality is linearly decodable at $t-1$ at three model scales, above token identity and every next-token uncertainty statistic, using a paired bootstrap over questions (Section 5.1).
- The scaling result: the probe improves with model size while uncertainty does not, so the advantage widens from $+0.112$ to $+0.204$ (Section 5.2).
- A demonstration that this comparison is fragile — sampling negatives from the whole rollout rather than the pivot-bearing span erases it (Section 5.3).
- A contrast between reading at $t-1$ and at the pivot, in which entropy inverts to below chance at all three scales (Section 5.4).
- The signed result: whether the impending shift is helpful or harmful is decodable at $t-1$ at 4B (0.780) though not at 0.6B, and is immune to both the uncertainty and token-identity accounts (Section 5.5).
- A characterization of two ways such headlines inflate: branch collapse and layer selection (Sections 3.1 and 5.6).

2 Related Work

Pivotal tokens. Abdin et al. [2024] introduce PTS to build preference pairs for Phi-4 post-training; Sharma [2025] released an open implementation and event datasets. Both use PTS as a data-generation step; the representational question—whether the property is present in activations beforehand—has not to our knowledge been asked, nor has the relationship between PTS labels and uncertainty been measured.

Uncertainty as the incumbent account. Wang et al. [2025] show that roughly 20% of tokens carry high entropy, act as branch points, and account for most of the gain from reinforcement learning with verifiable rewards. Sections 5.1 and 5.3 measure how much of PTS's criterion that account already covers, and how sensitive the measurement is.

Sentence-scale reasoning importance. Thought Anchors [Bogdan et al., 2025] identifies influential sentences by counterfactual resampling. We work at token scale, where the $t-1$ prediction problem is well posed: a sentence boundary is not a decoding step.

Probing and steering. Linear probes on intermediate representations [Alain and Bengio, 2018, Marks and Tegmark, 2023]; contrastive activation addition [Rimsky et al., 2024, Huang, 2024]; conditional steering [Lee et al., 2024]. Process reward models and value probes [Li et al., 2024] estimate $P(\text{success} \mid \text{prefix})$, a *level*; pivotality is $|\Delta p|$, a *sensitivity*.

3 Method

3.1 Generating the labels

We reimplement token-scale PTS with the same semantics as Sharma [2025] but express the bisection as a state machine, so that the rollouts of many questions can be batched together. The tree is sequential within a question—both children of a node need the node's own midpoint estimate—but questions are independent, and batching across them gives $8.9\times$ the generation throughput of running one node at a time. Emitted events carry the pivotal token's exact absolute position and the full searched rollout.

Two construction details materially affect the result.

Branches, not a canonical sequence. PTS samples several rollouts per question and they diverge after a few tokens. A construction that collapses a question to its longest rollout and matches the others against it as string prefixes silently discards every pivot on a diverging branch. On the released Qwen3-0.6B events, 104 questions carry 1,245 pivotal tokens over 447 distinct branches, and collapsing retains 318 of them—a 74% loss. We treat every maximal branch as its own sequence, and because branches share prefixes (and the residual stream at position p depends only on tokens $0 \ldots p$) we label each distinct context exactly once.

Positions are exact, not inferred. Byte-pair encoding does not guarantee $\mathrm{enc}(a+b) = \mathrm{enc}(a)+\mathrm{enc}(b)$, so recovering a pivot's index by re-tokenizing its prefix is unsafe: a pivot token such as "." frequently merges with the preceding word. For data we generate this problem does not arise. For the released datasets, which predate the position field, we verify each recovered index against the reported token id and discard rows that fail.

3.2 Label noise in the oracle

$\widehat{\Delta p}$ is a difference of two binomial estimates from S rollouts each, accepted when $|\widehat{\Delta p}| \geq \tau$. Under the null of no true change $\mathrm{Var}(\widehat{\Delta p}) = 2p(1-p)/S$, so at the reference setting $S{=}50$, $\tau{=}0.2$ the acceptance threshold sits at only 2σ: a false-acceptance rate near 4.6% per tested position at $p{=}0.5$. We therefore store the raw counts $(n_{\text{before}}, s_{\text{before}}, n_{\text{after}}, s_{\text{after}})$ rather than a verdict, which makes τ a post-hoc knob and lets every event carry a Wilson interval. [TODO: report the accuracy gap between the full label set and a high-confidence subset.]

3.3 Probes, baselines, and how they are compared

Probes are ℓ_2-regularized logistic regressions on the residual stream, fit on standardized features and converted back to raw activation space so the resulting vector is usable as a direction. Layer and regularization strength are chosen on an inner split of the *training* questions; the test split is used once.

Baselines: token identity, as a smoothed log-odds lookup and as one-hot logistic regression, both fit on train and scored on held-out rows; next-token entropy, negative top-2 margin, and negative top-1 probability, computed from the model's own distribution at $t-1$; and a random unit direction.

Comparisons are paired. Confidence intervals are bootstrapped over *questions*, since positions within one reasoning trace are not independent. Crucially, when comparing the probe against a baseline we bootstrap the *difference* of AUROCs on the same resampled questions, rather than checking whether two marginal intervals overlap. The latter is badly conservative and would, on our data, have supported a conclusion the paired test rejects.

4 Experimental Setup

Models: Qwen3-0.6B, Qwen3-1.7B and Qwen3-4B [Qwen Team, 2025] (28, 28 and 36 decoder layers; $d = 1024, 2048, 2560$). Task: GSM8K [Cobbe et al., 2021], conditioned on the bare question without a chat template, matching the released PTS datasets and therefore without extended-thinking mode. Search: 3,000 candidate questions per model, $S{=}40$ rollouts per probability estimate, one

Table 1: Data produced per model. "Qualifying" is the fraction of candidate questions whose baseline success probability fell inside the searchable band.

Model	Qualifying	Questions w/ events	Pivotal positions	Test questions	Test rows
Qwen3-0.6B	0.40	456	688	91	232
Qwen3-1.7B	0.45	520	835	104	287
Qwen3-4B	0.54	650	936	130	406

Table 2: Predicting whether the *next* token is pivotal, from the residual stream at $t-1$. Values are AUROC; parenthesised values are the paired difference (probe − baseline) with its 95% interval over question-level resamples. A random direction has null mean 0.49 with a 5–95 range of $[0.40, 0.62]$, so single draws in that band are unremarkable.

Detector	Qwen3-0.6B (L=22)	Qwen3-1.7B (L=23)	Qwen3-4B (L=24)
Linear probe	**0.807**	**0.863**	**0.897**
Mean-difference (CAA)	0.774 (+.034 [+.015, +.055])	0.841 (+.023 [+.008, +.037])	0.868 (+.029 [+.000, +.059])
Token identity (LR)	0.716 (+.091 [+.032, +.154])	0.715 (+.149 [+.105, +.191])	0.791 (+.107 [+.071, +.144])
Token identity (freq.)	0.696 (+.110 [+.050, +.169])	0.688 (+.176 [+.129, +.225])	0.785 (+.113 [+.078, +.150])
Neg. top-1 probability	0.704 (+.101 [+.003, +.201])	0.716 (+.148 [+.095, +.202])	0.700 (+.198 [+.136, +.264])
Neg. top-2 margin	0.699 (+.106 [+.006, +.207])	0.703 (+.161 [+.107, +.218])	0.703 (+.195 [+.134, +.259])
Next-token entropy	0.693 (+.112 [+.014, +.213])	0.721 (+.142 [+.090, +.195])	0.694 (+.204 [+.141, +.270])
Random direction	0.478	0.428	0.549

searched rollout per question, τ=0.2, 320 max new tokens, on a single Blackwell-class GPU per model.

PTS only visits questions whose baseline success probability lies inside $[0.2, 0.8]$: a question the model always or never solves has no pivotal token, because a saturated probability cannot move. Table 1 summarizes what each search produced. Notably the qualifying fraction *rises* with model size rather than falling — at 320 new tokens with raw conditioning, none of these models is close to saturating GSM8K.

Activations are read at every residual-stream position in bfloat16, at the labelled positions only. Negatives are sampled 1:1 from the span in which pivots occur (Section 5.3), and all splits are by question.

5 Results

5.1 Pivotality is decodable at $t-1$, at every scale

Table 2 reports each detector at each model scale, with the paired difference from the probe bootstrapped on the same resampled questions. The probe separates from every baseline at every scale.

5.2 The advantage over uncertainty grows with scale

The single most informative pattern in Table 2 is not any one number but a trend. Probe AUROC rises monotonically with model size, $0.807 \to 0.863 \to 0.897$, while next-token entropy stays essentially flat, $0.693 \to 0.721 \to 0.694$. The gap therefore widens from $+0.112$ (p=0.013) at 0.6B to $+0.204$ (p<0.001) at 4B.

This rules out the most deflationary reading of a single-scale result. If larger models were simply "easier to probe" in some generic way, the cheap detectors would improve alongside the probe; they do not. What improves is specifically the linear availability of pivotality in the residual stream, and what it is pulling away from is specifically predictive uncertainty.

Table 3: Probe read at $t-1$ versus at the pivotal token t, same data and same negatives. Entropy inverts to chance or below at all three scales.

	Qwen3-0.6B		Qwen3-1.7B		Qwen3-4B	
Detector	$t{-}1$	t	$t{-}1$	t	$t{-}1$	t
Linear probe	0.807	0.868	0.863	0.913	0.897	0.945
Mean-difference (CAA)	0.774	0.810	0.841	0.895	0.868	0.878
Token identity (freq.)	0.696	0.786	0.688	0.875	0.785	0.868
Next-token entropy	**0.693**	**0.449**	**0.721**	**0.475**	**0.694**	**0.495**
Neg. top-2 margin	0.699	0.445	0.703	0.454	0.703	0.494

The margin over *token identity* does not show the same trend ($+0.110 \to +0.149 \to +0.107$): token identity itself gets stronger with scale (0.696 $\to$ 0.715 $\to$ 0.791), so probe and lookup table improve together. The widening is uncertainty-specific.

5.3 Where the negatives come from decides the comparison

That comparison depends on a choice which is easy to leave implicit. Our rollouts run to the token limit, and the natural thing is to sample non-pivotal positions uniformly across the stored sequence. Doing so draws most of them from the fluent tail, *after* the model has committed to an answer: on Qwen3-0.6B, 63% of negatives landed beyond the last pivot, with mean next-token entropy 0.275 against 0.476 before it, and median position 232 against 146 for pivots.

Under that sampling entropy rises from 0.693 to 0.814 and the probe's advantage over it falls to $+0.039$ with an interval spanning zero — a null result manufactured entirely by the negative distribution. We therefore restrict negatives to the span in which pivots occur, and we recommend that any comparison of this kind state where its negatives came from. Published PTS datasets avoid the problem incidentally, because the sequence they store stops at the deepest pivot; it appears as soon as one keeps the full rollout, which is otherwise the more useful artifact.

5.4 Reading at the pivot instead of before it

Moving the label from $t-1$ to the pivot itself gives a post-hoc classifier: informative, but unable to gate anything.

The probe and token identity both rise, which is unsurprising: at t the token occupying the labelled position *is* the pivot. The informative change is entropy, which inverts from ≈ 0.70 to ≈ 0.47 at all three scales. Uncertainty is elevated immediately before a pivotal token and suppressed immediately after it — what a decision point being resolved should look like. The two positions carry opposite uncertainty signatures rather than differing in degree, and this supplies an argument for the $t-1$ framing that does not depend on the intervention story: $t-1$ is where the uncertainty signal lives.

We do not report the t column as a headline. Its null is noisier — a random direction reaches 0.55–0.68 there against 0.43–0.55 at $t-1$, because the classes are more separable overall, so a single random projection picks up more of that separation by chance.

5.5 The sign of the impending shift is decodable, at scale

Pivotality as defined in Eq. 1 is symmetric: a token that rescues an answer and one that ruins it receive the same label. The sharper question is whether the *sign* is decodable at $t-1$ — whether the model represents, one step early, that it is about to make a mistake.

This question is immune to Section 5.3 by construction. Entropy is sign-blind: it cannot distinguish a helpful branch point from a harmful one, and indeed scores *below* chance on this task at every scale (0.514, 0.428, 0.434). Whatever a signed probe reads, it is not predictive uncertainty.

Restricting to pivotal positions and relabelling by $\mathrm{sign}(\Delta p)$ gives Table 4.

At 0.6B this is a clean null: the probe separates from nothing, not even a random direction. At 1.7B it clears uncertainty and the random null but not token identity. At 4B it separates from every

Table 4: Helpful versus harmful, from the residual stream at $t-1$. Parenthesised values are the paired difference from the probe with its 95% interval. The task goes from a clean null at 0.6B to separated from every cheaper explanation at 4B.

Detector	Qwen3-0.6B	Qwen3-1.7B	Qwen3-4B
Signed probe	**0.614**	**0.650**	**0.780**
Mean-difference (CAA)	0.567 (+.046, ns)	0.595 (+.056, ns)	0.757 (+.021, ns)
Token identity (freq.)	0.555 (+.057, ns)	0.610 (+.037, ns)	0.691 (+.089 [+.026, +.151])
Next-token entropy	0.514 (+.098, ns)	0.428 (+.219 [+.085, +.348])	0.434 (+.346 [+.237, +.452])
Random direction	0.545 (+.065, ns)	0.498 (+.153 [+.050, +.254])	0.505 (+.273 [+.164, +.375])

cheaper explanation (+0.089 over token identity, p=0.002). The only baseline it does not clear at 4B is the mean-difference direction (0.757, p=0.160), and that closeness is informative rather than deflationary: it says the signal is essentially a contrastive direction, which is precisely the object one would use for an intervention.

An earlier version of this analysis, on a smaller and noisier label set at 0.6B alone, gave 0.717 and looked promising; it did not survive a cleaner dataset. The result reported here is the opposite case — weak where the data is weak, and strong only where scale supports it.

Why $t-1$ is the honest position for this question. Reading at the pivot instead raises the signed probe to 0.697, 0.859 and 0.890. But most of that is not representation: token identity rises from 0.56–0.70 at $t-1$ to 0.74–0.85 at t, because at the pivot the token being classified *is* the pivot, and helpful and harmful tokens differ lexically. At 0.6B the lookup table actually beats the probe there (0.735 versus 0.697). At $t-1$ that escape route is closed by construction — a lookup on the current token cannot encode the sign of a token that has not been emitted — so the $t-1$ column is the one that isolates representation from vocabulary.

5.6 No layer is special, and selection inflates the headline

On Qwen3-0.6B, per-layer accuracy spans 0.651–0.741 with mean 0.702. Simulating a null in which all 29 layers are equally informative, each measured on 232 rows, gives an expected maximum of 0.762 against an observed 0.741. There is no evidence of layer specialization; the signal is spread across the mid-to-late stack, and naming a single "pivotal layer" would be an artifact of maximizing over correlated noise. Consistently, the selected layer sits at 76–79% of depth in all three models (L=22/29, 23/29, 24/37) without any of them being distinguishable from its neighbours.

The cost of getting the protocol wrong is measurable. Choosing the layer on training data and reporting on test gives 0.807; taking the best layer on the reporting split gives 0.814. Under a leakier protocol we measured the same gap at 0.063 — large enough to manufacture a result.

5.7 Regularization

On Qwen3-0.6B, AUROC moves from 0.752 at C=1 to 0.816 at C=10^{-3}, and the probe direction rotates toward the mean-difference direction as the penalty tightens (cosine 0.173 to 0.765). A weakly regularized probe still classifies, but along a substantially different direction — which matters for anyone intending to use the weights as an intervention direction rather than only as a classifier.

6 Limitations

A single 0.6B model on a single task family, with 91 held-out questions. The oracle's labels are themselves noisy (Section 3.2) in a way we have quantified analytically but not yet empirically. The acceptance filter retains only mid-difficulty questions, so pivot density is confounded with task difficulty relative to model capability and this label set is not a uniform sample of GSM8K. Conditioning on the bare question puts an instruct model off distribution; we observed rollouts in which the model hallucinated multiple-choice options, and at least one event whose extreme Δp traces to the answer-extraction cascade matching such a hallucination rather than to a genuine shift.

Linear probes test linear decodability only. Everything here is correlational: we have not shown that the direction the probe identifies is causally used.

Most importantly, the margins over uncertainty in Table 2 are $+0.10$ to $+0.11$ with lower bounds within 0.01 of zero, on 91 questions. They are separations, not comfortable ones, and Section 5.3 shows how little it takes to erase them. [TODO: Scale the question count and re-test; extend to Qwen3-1.7B and 4B.]

7 Conclusion

Pivotality is linearly present in a model's residual stream one position before the token that realizes it, at 0.807 to 0.897 AUROC across three model scales, above token identity and above every next-token uncertainty statistic we tested. The advantage over uncertainty widens with scale while uncertainty itself stays flat, which is the strongest evidence we have that the probe reads something other than local predictive difficulty. Reading at the pivot instead inverts entropy to below chance at every scale, locating the uncertainty signal specifically at $t-1$ — the one position that is both informative and available while a decision can still be influenced. The *direction* of the impending shift is decodable too, but only at scale: a null at 0.6B and 0.780 AUROC at 4B, separated from token identity, from uncertainty (which is sign-blind and scores below chance), and from a random direction. That the signed signal is closely approximated by a contrastive mean-difference direction makes it the natural object for a causal test, which is the next step.

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A Reproduction

Events: `stvngo/Qwen3-0.6B-pts-tokens`. Activations: `stvngo/Qwen3-0.6B-pivotal-activations`. All numbers in Section 5 come from `artifacts/probes_v2_full/qwen3-0.6b_report.json`, produced by `scripts/pts_run.py`, `scripts/build_and_extract.py` and `scripts/evaluate_probes_v2.py`. [TODO: Add the layer sweep and regularization-path figures.]